Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/18/2025
PI Name:	Vera, Alonso Ph.D.		
Project Title:	Assessment of the State of Communication Delay Research in Preparation for Missions Beyond Low Earth Orbit		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (3) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/05/2023	**End Date:**	09/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2024 per D. Kulkarni/NASA HRP. Original end date was 04/05/2024 (Ed., 10/30/24).		
Key Personnel Changes/Previous PI:	NOTE: Per the Human Factors and Behavioral Performance (HFBP) Element, the Principal Investigator for this project changed from Lauren Landon, Ph.D. to Alonso Vera, Ph.D. (Ed., 2/18/2025).		
COI Name (Institution):	Panontin, Tina Ph.D. (San Jose State University Research Foundation Inc) Landon, Lauren Ph.D. (KBR Wyle/NASA Johnson Space Center) Karasinski, John Ph.D. (NASA Ames Research Center) McTigue, Kaitlin Ph.D. (NASA Ames Research Center) Parisi, Megan (NASA Ames Research Center) Wu, Shu-Chieh Ph.D. (San Jose State University Research Foundation Inc)		
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Task Description:	NASA's mission-operations paradigm, established during Project Mercury and minimally evolving through the Apollo Program, Space Shuttle Program, and ISS missions, has largely depended on real-time support from a ground team of experts. This ground team has served as the safety net for crewed spaceflight missions over the past 60 years, managing the combined state of the mission, vehicle, and crew. However, this operational paradigm, which has seen little change in its Human-Systems Integration Architecture (HSIA), will face challenges during long-duration exploration missions beyond low Earth orbit. In deep-space crewed missions, the crew will be confronted with infrequent resupply opportunities, no viable options for evacuation or rescue, and high-latency communications, prohibiting real-time operational and medical support. Due to the great distances from Earth and communication architectures involved, lunar missions may experience one-way communication latencies ranging from 3 to 14 seconds, while Mars missions will encounter up to 44 minute round-trip delay at maximum distance from Earth. NASA's Human Systems Risk Board has identified and assessed the risk associated with reduced ground support and the necessary shift towards more Earth-independent operations. Retrospective analyses of spaceflight anomalies reveal a persistent and significant rate of unanticipated anomaly occurrence (e.g., 1.7/year for the International Space Station / ISS). Qualitative causal modeling using a Directed Acyclic Graph (DAG) was employed to assess the consequences of this risk and showed that catastrophic outcomes, such as Loss of Crew (LOC), are possible when onboard support (HSIA) is insufficient to mitigate the reduced ground support as the distance from Earth increases. Given the high likelihood of the HSIA risk and its severe consequences, it is assessed as "red" for extended missions on the lunar surface and Mars. Understanding the impact of communication delays on executing complex operations and problem-solving is key to grasping the magnitude and mitigation of this risk. Real-time communication allows the crew to rely on a large, extensively resourced ground team to oversee and direct operations and diagnose and resolve issues. However, as communication delays increase, these responsibilities must increasingly shift to the onboard team. Incremental increases in communication delay will require the onboard crew to gradually assume roles historically filled by the ground team. NASA has investigated the nature and effectiveness of managing this shift in complex operations since early 2000. Key findings to date include: • Even short communication delays can disrupt interactions between Mission Control Center (MCC) and the crew. • As transmission delays increase, space-ground communications degrade significantly. • Degraded communications adversely affect team performance. • Countermeasures for asynchronous communication require more study. NASA has also studied the impacts of communication delays on the behavioral health and performance of individuals and crews operating across the multi-team space-to-ground system. Previous research indicates increased isolation, and the challenges posed by delayed communication lead to heightened stress, frustration, adverse behavioral symptoms, and reduced individual performance. Individuals experience increased stress and reduced effectiveness of countermeasures, including Private Psychological Conferences (PPCs) and Private Family Conferences (PFCs). Additionally, family connectedness and satisfaction with interpersonal relationships decline in such environments. At the team and multi-team levels, communication delays undermine shared understanding, team coordination, performance, and cohesion between space and ground teams. The crew experiences greater isolation, compounded by an increased workload and interpersonal friction within the high-tempo environment. Despite this substantial body of work, additional research is necessary to comprehensively assess the issues associated with communication latencies and identify effective countermeasures. The purpose of this work is to examine the evidence from 20 years of research on communication delays, integrating findings across Behavioral Medicine (BMed), Team, and HSIA Risks to develop a coherent understanding of what has been studied, how it has been approached, and the outcomes measured. Additional perspectives are gained from interviews with the research participants and researchers themselves. Preliminary results indicate: • Gaps exist in research on short lunar-like delays due to the historical prioritization of Mars-like delays in NASA research initiatives. • ISS and analog (e.g., the Human Exploration Research Analog / HERA) safety requirements and mission objectives limit studies of communication delay impacts to mission-critical operations. • Gaps exist in research on communication delay breakpoints.
Rationale for HRP Directed Research:	This proposal qualifies for a directed study due the highly constrained nature of the research. The project requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal. The research activity involves analog and operations practices and associated personnel and includes data mining studies.
Research Impact/Earth Benefits:	This work informs research gaps around specific combinations of amount of comm delay, tasks to be performed, and metrics. As such the outcomes are relevant to any terrestrial context with comm denied/delayed conditions and/or reduced expert support when distributed teams are involved.
Task Progress:	A full assessment of the literature on the impacts of communication delay delay on distributed performance has been completed. On missions beyond low Earth orbit (LEO), communication delays will challenge the traditional mission operations paradigm, reducing ground support and increasing crew autonomy. Communication delays have been known to negatively impact crew behavioral health and performance and to affect the coordination and cohesion of the space-to-ground team. However, research remains to be done to comprehensively assess the issues associated with communication delays and what is needed to overcome them. Our literature review surveyed evidence generated through 20 years of communication delay research to build a picture of what has been studied and how, and to identify gaps in the evidence base that threaten upcoming mission priorities. The gap in studies of short communication delays (i.e., 3 to 14 seconds one-way) possible on lunar missions is of particular interest. We also interviewed subject matter experts, including Flight Operations Personnel, former analog (e.g., HERA, CHAPEA) crewmembers, and researchers who study communication delay to identify knowledge gaps and research priorities for upcoming missions. [Ed. Note: HERA is the NASA Human Exploration Research Analog. CHAPEA is the NASA Crew Health and Performance Exploration Analog.] Our analysis reveals a general lack of fidelity in studies of mission-critical tasks, including problem solving, emergency response, maintenance, and complex procedure execution—likely due to the prioritization of safety and mission objectives in analog missions. There remains a pressing need to study the impact of different delays in combination with specific tasks, contexts, and environments to adequately characterize the risk of reduced

	ground support for missions to the Moon and Mars. Countermeasures, including training specific to communication delay and tools to facilitate asynchronous collaboration, may mitigate the impact of communication delay and need to be designed and evaluated for specific contexts (i.e., chatting with family vs. getting a briefing from the ground on a maintenance procedure).
Bibliography Type:	Description: (Last Updated: 02/18/2025)
Abstracts for Journals and Proceedings	Parisi ME, McTigue KR, Wu S-C, Karasinski JA, Panontin TL, Landon L, Vera AH. "The impact of delayed communication on NASA's Human-Systems operations: Preliminary results of a systematic review." 15th International Conference on Applied Human Factors and Ergonomics (AHFE 2024), Nice, France, July 22-27, 2024. Abstracts. 15th International Conference on Applied Human Factors and Ergonomics (AHFE 2024), Nice, France, July 22-27, 2024. , Jul-2024